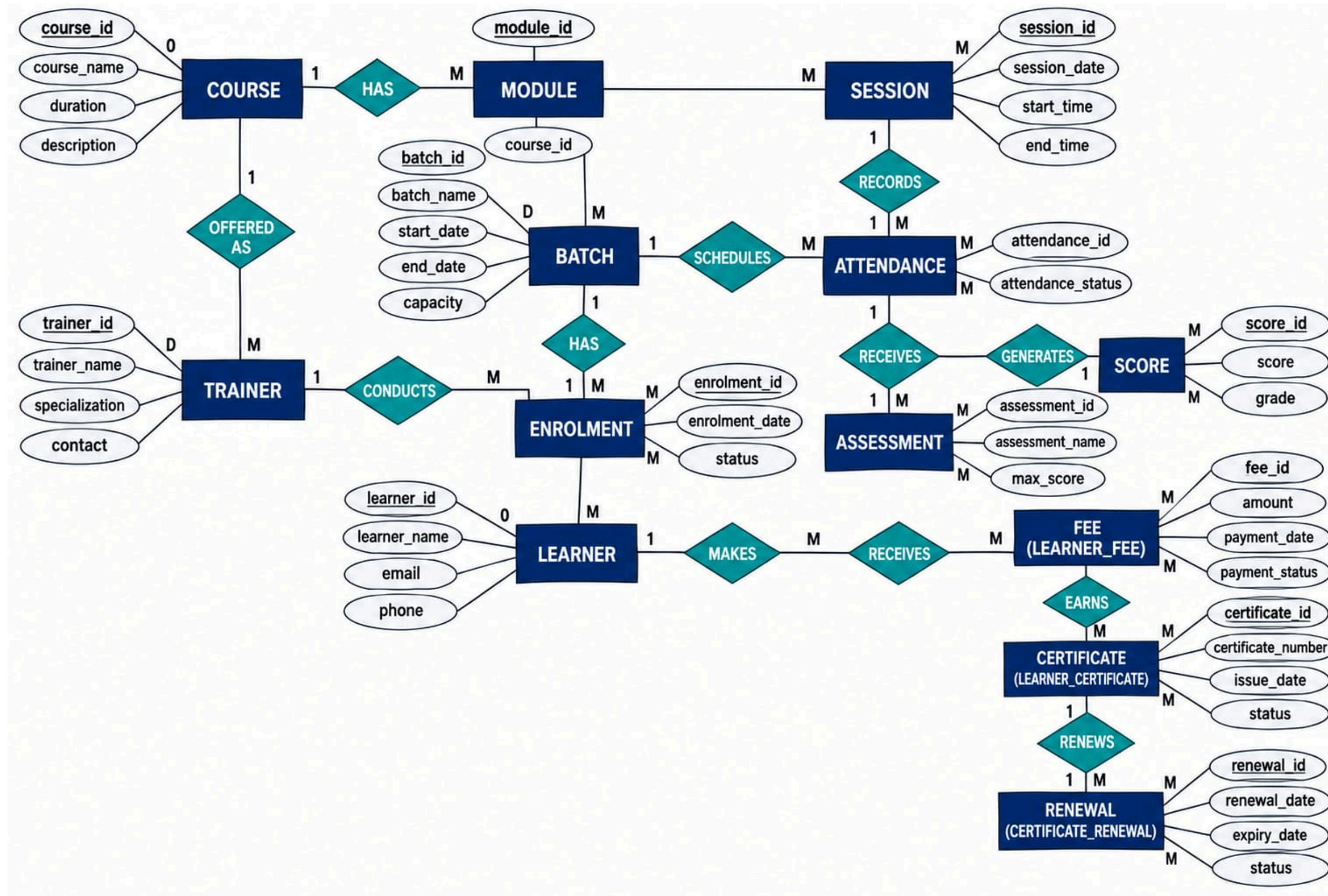


PROJECT 33

Design and Implementation of a Database Management System for Training Programme

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Conceptual Design — ER diagram



How the Entities Are Connected

The ER diagram in plain words — every link is one-to-many (1:M).

A Course *has many* Modules and Batches

A Learner *can have many* Enrolments

A Session *belongs to* one Batch and one Module

A Batch *has many* Assessments

An Enrolment *produces one* Fee record

A Certificate *may have many* Renewals

A Trainer *is assigned to* many Batches

A Batch *has many* Enrolments and Sessions

An Enrolment *leads to many* Attendance records

An Assessment *produces many* Scores

An Enrolment *can lead to one* Certificate

Initial Relational Schema

Proposed tables, primary keys and how they relate.

Course — course_id, course_name, duration, description
Module — module_id, module_name, course_id
Trainer — trainer_id, trainer_name, specialization, contact
Batch — batch_id, batch_name, start_date, end_date, capacity, course_id, trainer_id
Learner — learner_id, learner_name, email, phone
Enrolment — enrolment_id, enrolment_date, status, learner_id, batch_id
Session — session_id, session_date, start_time, end_time, batch_id, module_id
Assessment — assessment_id, assessment_name, max_score, batch_id, module_id
Attendance — attendance_id, attendance_status, session_id, enrolment_id
Score — score_id, score, grade, assessment_id, enrolment_id
Fee — fee_id, amount, payment_date, payment_status, enrolment_id
Certificate — certificate_id, certificate_number, issue_date, status, enrolment_id
Renewal — renewal_id, renewal_date, expiry_date, status, certificate_id

What has been prepared so far.

Summary — Ready for Review 2

1

Problem, scope, objectives, users and functional requirements are documented.

2

The ER diagram shows 13 entities and how they connect.

3

An initial relational schema proposes tables, primary keys and relationships.

Next Step

After Review 1, we will build the actual database, add sample data, write SQL queries and reports, and connect the front-end app.